Lecture 40: Electric Machines

Induction machines...

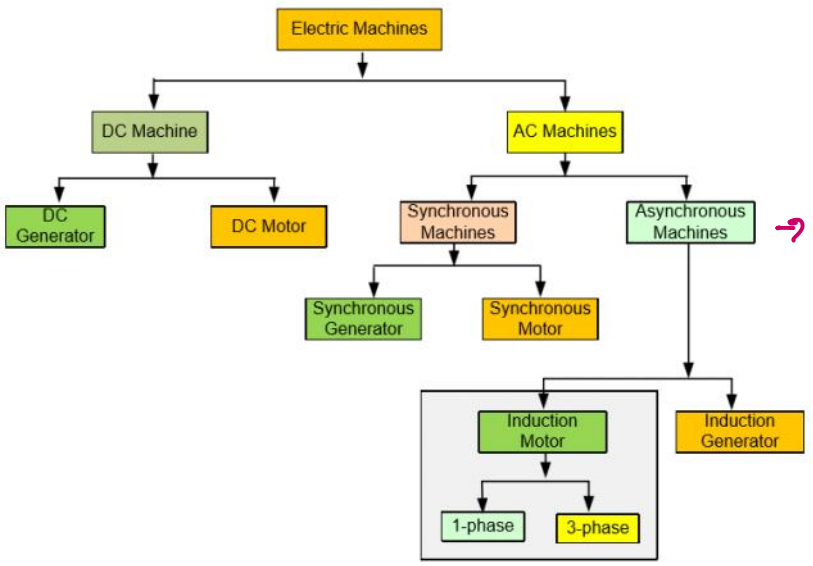
Basic Electrical Engineering (EE1100)

INDUCTION MACHINES

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Indian Institute of Technology Madras
July-Nov. 2017

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Electric Machine Classification



→ Induction m/c

Applications of Electric Machines

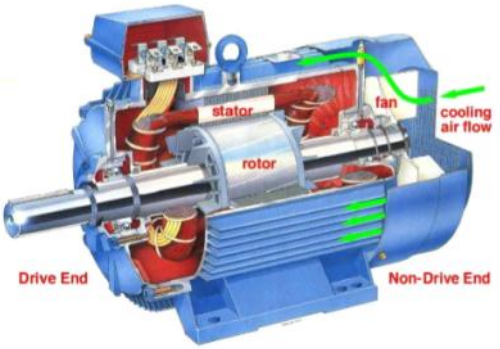
Type of Machine	Common Uses / Applications
DC Generator	Used in electroplating, battery charging, DC welding, and excitation of synchronous machines.
DC Motor	Employed in electric traction (trains, trolleys), rolling mills, cranes,

	elevators, and variable-speed drives.
Synchronous Generator	Acts as the main source of electric power in thermal, hydro, and nuclear power stations.
Synchronous Motor	Used in industries requiring constant speed — compressors, pumps, conveyors, and power factor correction.
Induction Motor (1-phase)	Operates small domestic appliances — fans, mixers, washing machines, and blowers.
Induction Motor (3-phase)	Drives industrial machinery — pumps, compressors, conveyors, and lathes.
Induction Generator	Serves in wind turbines, micro-hydel plants, and regenerative braking systems.

Introduction

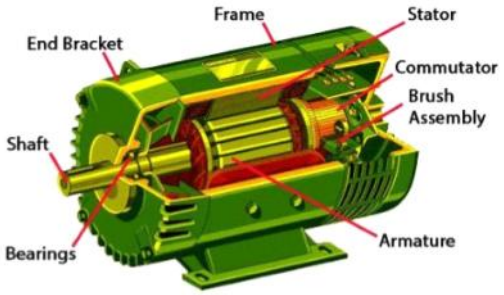
Types of Induction Machines

- 3-phase induction machines
  1) Squirrel cage induction motor
  2) Slip-ring induction motor or wound rotor induction motor
- 1-phase induction machines
  1) Split phase induction motor
  2) Capacitor start induction motor



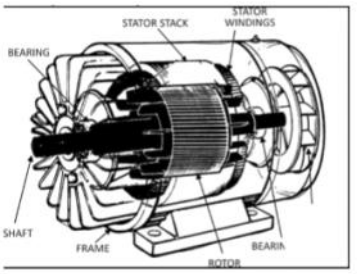
Cross sectional view of an induction machine

Induction Motor

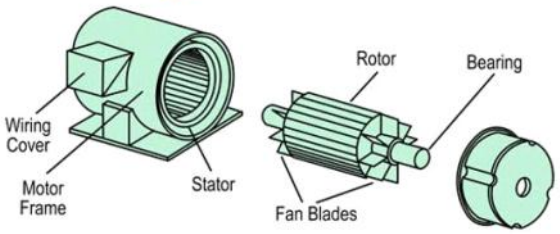




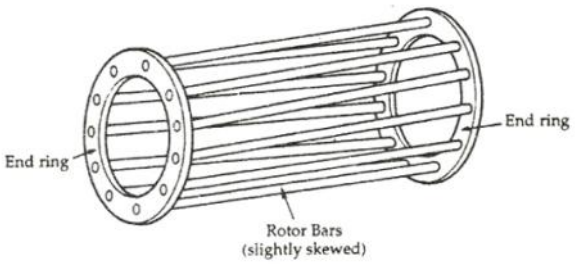




Squirrel Cage Induction Motor



Basic construction of squirrel cage induction motor



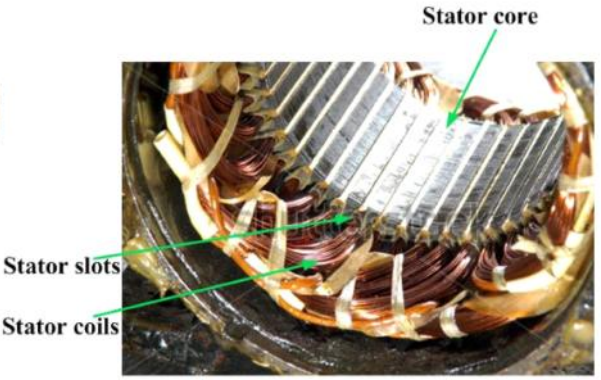


Squirrel cage rotor (Wikipedia)

Skewing: The **rotor** bars are **skewed** in Induction Motor. This is done to prevent Cogging. This happens when the **rotor** slots and stator slots are same in number or they are integer multiples of each other. due to this the opposite poles of stator and **rotor** come in front of each other and get locked.

Stator of an Induction Motor

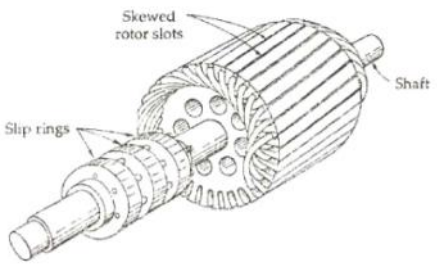




Stator windings in an induction motor

- Stator core is laminated in order to reduce the eddy current losses and it is generally made of silicon steel in order to reduce hysteresis loss.
- Slots on the periphery of stator core is where the windings are distributed and wound in series.

Slip Ring Induction Motor

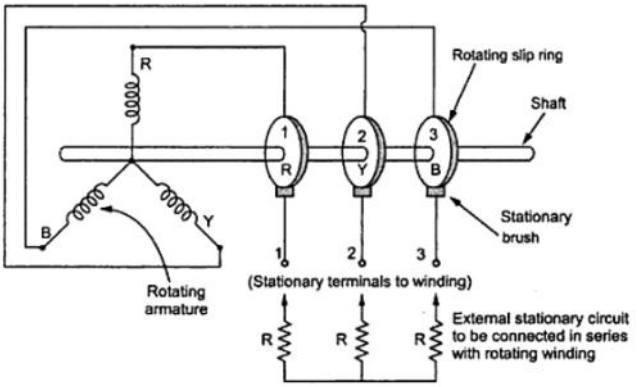




Slip ring rotor construction

- The rotor is wound for the same number of poles as that of stator, but it has lesser number of slots and lesser turns per phase.
- Rotor coil is permanently connected to the slip rings.
- External resistance or external circuitry can be connected to rotor through brushes assembly on the slip ring.

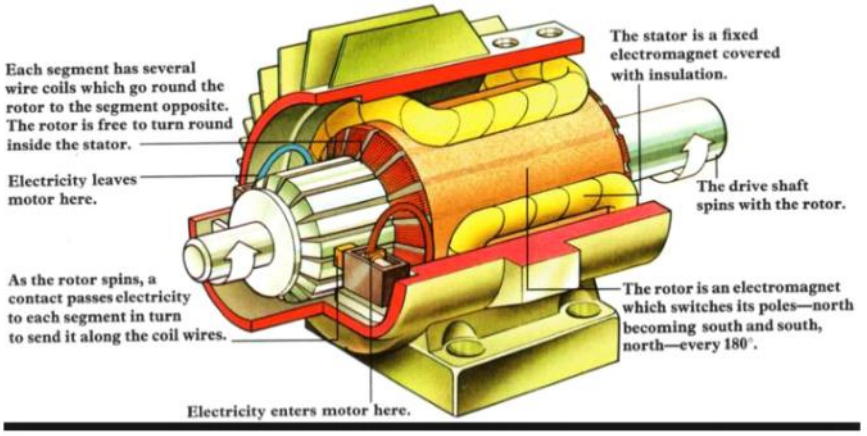
Contd..



Slip ring rotor representation

- External resistance is added to the rotor windings either during starting of induction motor or for speed control of induction motor.
- Other electrical circuits may also be added in order to achieve different functions like slip recovery scheme.

Operation of induction motor



Merits and Demerits of Squirrel Cage Induction Motor

Advantages:

- Squirrel cage induction motors are cheaper in cost and much more rugged in construction compared to slip ring induction motors.
- Due to absence of slip ring and brushes, maintenance duration and cost associated with wear and tear of brushes is minimized.
- Compared to dc and synchronous motors, the absence of commutator, slip rings and brushes maintenance is minimized.
- DC motors when operated in polluted and hazardous environment (e.g. mining) has probability of fire due to sparking in brushes which are absent in squirrel cage induction motors.
- 3-phase induction motors have self starting torques unlike synchronous motors, hence neither squirrel cage nor slip ring induction motor require any additional circuitry for starting.

Contd..

Disadvantages of squirrel cage induction motors:

- Squirrel cage induction motors have a low starting torque and high starting currents. Thus, in order to limit starting current methods like star-delta starter, auto transformer starting, etc. are required.
- During voltages surges, increase in voltage saturates magnetic components of squirrel cage induction motor and during voltage sag induction motor draws more current.
- Induction motor operate under lagging power factor and during light load conditions they operate at low power factors. Hence, power factor correction equipment is needed near these motors.
- The precision of speed control in induction motors is low when compared to that of dc motors.
- The methods of obtaining wide range of speed control in induction motor is complicated when compared to dc motors.

Merits and Demerits of Slip Ring Induction Motor

Advantages:

- External resistance can be added to the rotor windings in order to reduce the large starting currents in stator winding and provide higher starting torques.
- By varying external resistance added to rotor, speed control of slip ring induction motor is possible.
- Slip power recovery scheme can be used in rotor winding to control speed in an energy efficient method.
- Slip ring machines are highly suited for use of generators in wind turbine systems.

Disadvantages:

- Due to slip ring and brush assembly the maintenance required is higher and it may cause sparking.
- Slip ring induction motors are sensitive to fluctuations in supply voltage.

Types of Squirrel Cage Induction Motor

- NEMA in US and IEC in Europe has classified the design of squirrel cage induction motors based on speed-torque characteristics and stator current rating into different classes.

  Class A - Normal starting torque, normal starting current and normal slip

  Class B - Normal starting torque, low starting current and normal slip

  Class C - High starting torque, low starting current and normal slip

  Class D - High starting torque, low starting current and high slip

  Class E - Low starting torque, normal starting current and low slip

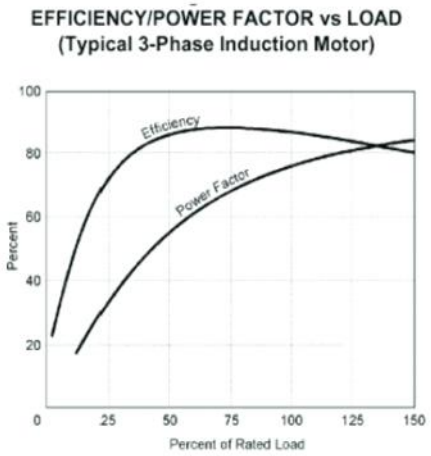
- Slip here implies the slip of motor corresponding to the rated speed.
- Class A motors are the popular type of induction motor and used in fans, compressors, conveyors applications which are having low inertia loads.

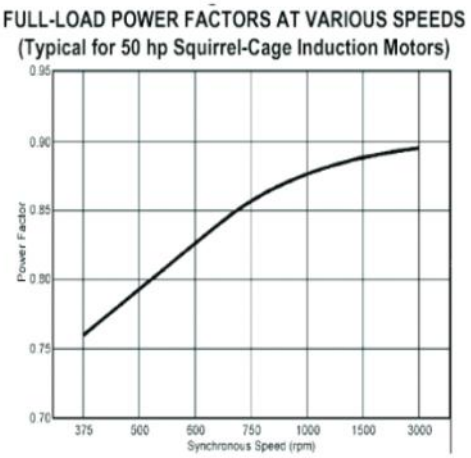
NEMA: National Electrical Manufacturers Association
The Association of Electrical Equipment and Medical Imaging Manufacturers

IEC: International Electrotechnical Commission (Formed in 1906, UK)

Parameters related to Induction Motor

- Two of the most important parameters relating to induction motors while selecting for an application are efficiency and power factor.
- Squirrel cage motors are normally more efficient than slip-ring motors, and higher-speed motors are normally more efficient than lower-speed motors.

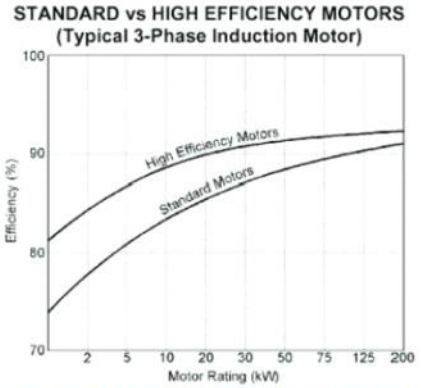




Reference: Bureau of Energy Efficiency India

Contd..

- Energy-efficient motors are designed with focus on reducing intrinsic motor losses.
- Improvements includes use of lower-loss silicon steel, thinner laminations of stator core, smaller air gap between stator and rotor, copper instead of aluminum bars in rotor, better ventilation, etc.
- But, the costs of energy-efficient motors is higher compared to standard induction motors.



Reference: Bureau of Energy Efficiency India

Applications of Induction Motor

Application in cranes





Application in Excavators



Applications of Induction Motor Contd.,

Applications in Drill-Press



Applications in Power Tools



Application in High speed vacuum cleaners





Applications of Induction Motor Contd.,

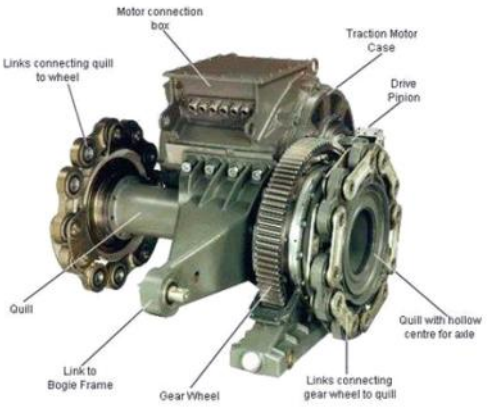
Application in Electrical Vehicles



Fig. The Tesla Roadster was an early adopter of the new generation of advanced AC induction motors that offer an alternative to permanent magnet motors which use rare earth materials in their construction.

Applications of Induction Motor Contd.,

Traction Applications





Applications of Induction Motor Contd.,

Application in Irrigation Pumps





Applications of Induction Motor Contd.,

Application in Thermal Power Plants



Fig. Induced draft (ID) fan used for collecting hot flue gases from furnace and delivering to chimney.



Fig. Forced draft (FD) fan used for providing the required quantity of hot air to the furnace for smooth and uniform combustion of fuel.

Applications of Induction Motor Contd.,

Application in lathes and grinding tools



Lathe applications



Grinding tool

Applications of Induction Motor Contd.,

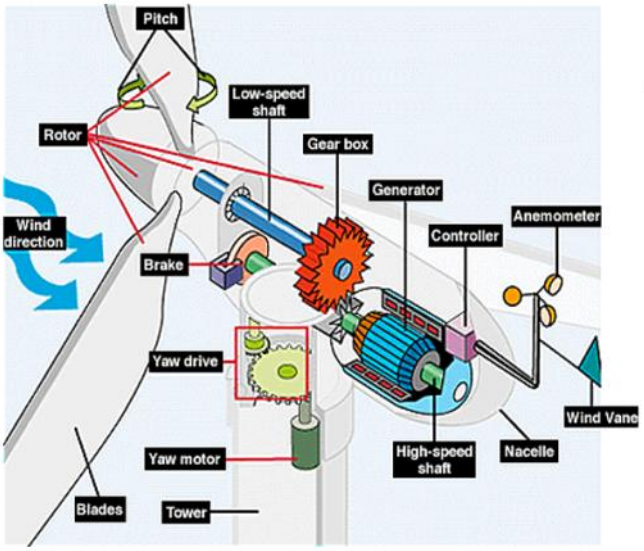
Application in Industrial pumps



Induction motor used for pumping of coolants, air etc. in industrial plants

Applications of Induction Motor Contd.,

Applications in Wind Power Generation

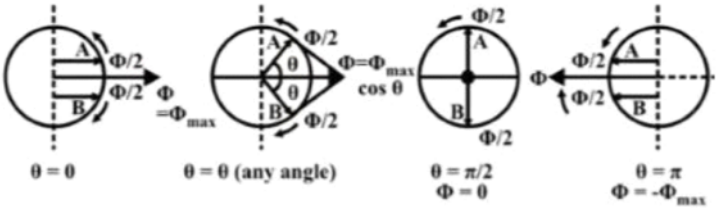


- Induction machines are generally used in wind power plants as generators
- Induction generators are capable of rotating at different speeds unlike the conventional ac generator (synchronous generator) and is the best choice for wind power plants



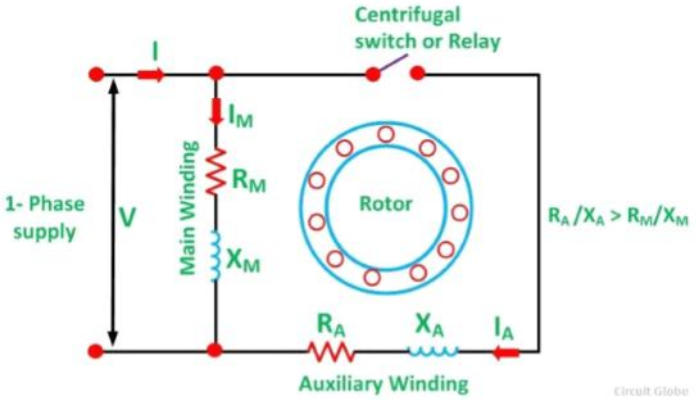
Single Phase Induction Motor

- A single phase induction motor has only one distributed stator winding and a squirrel cage rotor.
- When fed with single phase supply, stator produces a flux alternating in only single space axis which can be explained with help of double field revolving theory.



- By the double field revolving theory stator flux can be decomposed a forward component and backward component which revolve in opposite directions.
- Also, since these components are equal, they cancel each other and the net starting torque experienced by the rotor is zero.
- Thus, single phase induction motor is not a self-starting motor.

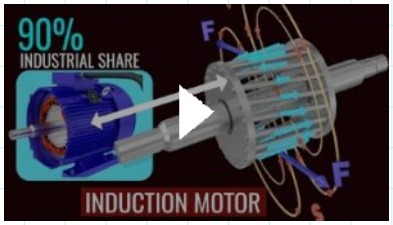
Split Phase Induction Motor



- The main winding and auxiliary winding are designed according to the relation shown in figure.
- Hence, current in two windings have phase difference of 25^0 to 30^0.
- As the motor behaves like a two phase motor, revolving flux is developed and hence motor is self starting.
- The centrifugal switch is opened when motor almost reaches its rated speed.

Check the following links for animations to develop more clarity on operation and working of Induction Machines

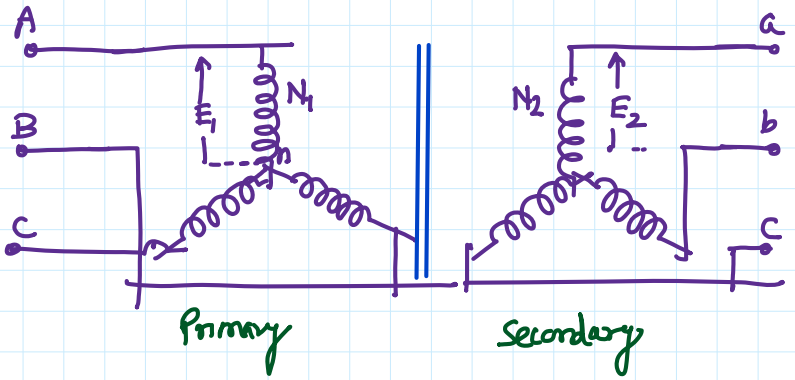
1. **How does an Induction Motor work?**
DON'T SUBSCRIBE | How does an Induction Motor work?



2. Three-**Phase Induction Motor**
https://www.youtube.com/watch?v=wzkH2pGWy4q

https://drive.google.com/drive/folders/10zzQZLRluzCYM_w5dRu4bFU7rMxllnCQ?usp=sharing

Induction Machine as a Transformer



$$E_1 = 4.44\, f_1\, N_1'\, \phi_{max}$$
$$E_2 = 4.44\, f_2 \cdot N_2'\, \phi_{max}$$

$N_1' = K_{w1} * N_1$

$N_2' = K_{w2} * N_2$

ϕ_{max} = Maximum flux in the core

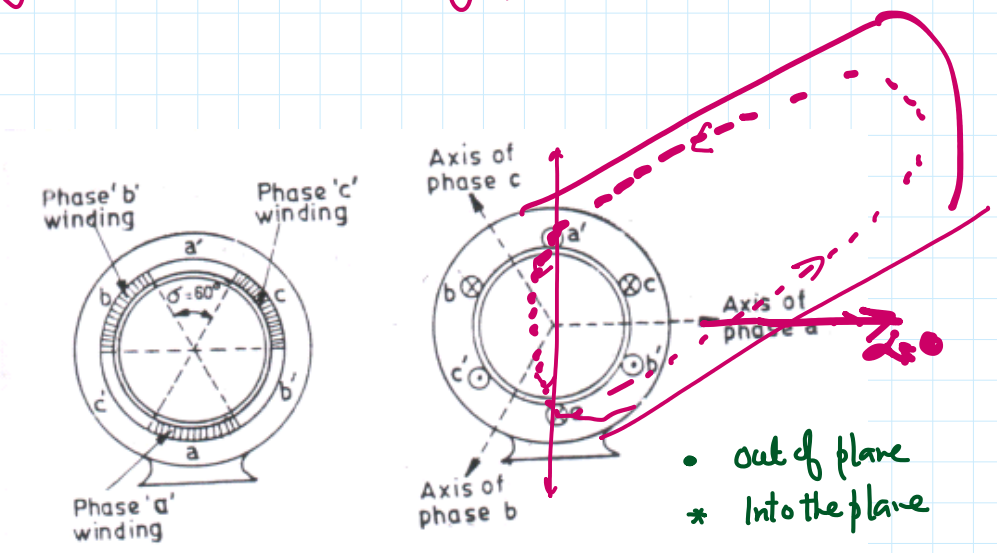
At stand still $\Rightarrow$

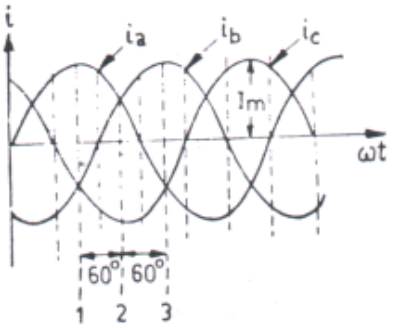
$$\boxed{\frac{E_1}{E_2} = \frac{N_1'}{N_2'}} \quad ; \quad \frac{E_1}{E_2} = \frac{f_1 N_1'}{f_2 N_2'} \quad ;$$

The only difference in transformer and induction machine has rotating part. There is airgap between the rotor and stator, which has more reluctance or more leckage of flux, hence I_o = No load component of current is more $\simeq$ 30-50% of I rated against 2-6% for transformer.

for transformer.

Revolving Magnetic Field in the Airgap:





$$\begin{cases} v_a = V_m \sin \omega t \\ v_b = V_m \sin(\omega t - 120^\circ) \\ v_c = V_m \sin(\omega t + 120^\circ) \end{cases}$$

$$\hookrightarrow \begin{cases} i_a = I_m \sin(\omega t - \phi) \\ i_b = I_m \sin(\omega t - 120^\circ - \phi) \\ i_c = I_m \sin(\omega t + 120^\circ - \phi) \end{cases}$$

$$\left.\begin{array}{l} \text{Amp Turns}_a = N i_a \\ \text{Amp. Turns}_b = N i_b \\ \text{Amp Turns}_c = N i_c \end{array}\right\} \begin{array}{l} = N' I_m \sin(\omega t - \phi) \\ = N' I_m \sin(\omega t - 120^\circ - \phi) \\ = N' I_m \sin(\omega t + 120^\circ - \phi) \end{array}$$

$$N\frac{d\phi}{dt} = L\frac{di}{dt} \quad \text{or} \quad N(\phi) = L(i)$$

$$\uparrow NL$$

$$\begin{cases} F_a(\alpha, t) = N i_a \cos\alpha = N I_m \sin\omega t \cdot \cos\alpha \\ F_b(\alpha, t) = N i_b \cos(\alpha - 120^\circ) = N I_m \sin(\omega t - 120^\circ)\cos(\alpha - 120^\circ) \\ F_c(\alpha, t) = N i_c \cos(\alpha + 120^\circ) = N I_m \sin(\omega t + 120^\circ)\cos(\alpha + 120^\circ) \\ F_R(\alpha, t) = F_a(\alpha, t) + F_b(\alpha, t) + F_c(\alpha, t) \end{cases}$$

